IS101 — Introduction to Computing

Client Billing Workbook

Desert Sun Consulting · Unit 3 — Spreadsheets · Core Project 1 · 100 points · 25 tasks

How to use this document: Download the starter workbook AND the clients.csv file from the Project Grader page. Open the starter in Microsoft Excel 365 and keep this sheet open beside it. Work through each task in order. When you're done, save the file and upload it to the grader for instant scoring.

D1 — Workbook Setup 25 points · 7 tasks

Task 1 5 points

Import the client roster from clients.csv

WHY THIS MATTERS

Real-world data often arrives as CSV files that need to be imported into Excel for analysis.

WHERE TO FIND IT

Click on the Clients sheet tab → Data tab → Get Data group → From Text/CSV → browse to the clients.csv file you downloaded → Import → Load.

HOW YOU'LL KNOW IT WORKED

The Clients sheet shows client data in columns: Client ID, Client Name, Primary Contact, and Region.

Task 2 3 points

Set the Billing sheet to Landscape orientation

WHY THIS MATTERS

The billing data has many columns — landscape orientation fits more columns on a printed page.

WHERE TO FIND IT

Click the Billing sheet tab → Page Layout tab → Orientation → Landscape.

HOW YOU'LL KNOW IT WORKED

The page is wider than it is tall.

Task 3 4 points

Add a header with company name and a footer with page numbers

WHY THIS MATTERS

Headers and footers identify printed pages and add page numbers for multi-page workbooks.

WHERE TO FIND IT

Page Layout tab → Page Setup dialog launcher (small arrow) → Header/Footer tab → Custom Header: type “Desert Sun Consulting” → Custom Footer: insert page number.

HOW YOU'LL KNOW IT WORKED

Print Preview shows the company name at the top and page numbers at the bottom.

Task 4 3 points

Freeze the top header row

WHY THIS MATTERS

Freezing keeps column headers visible while scrolling through data rows.

WHERE TO FIND IT

Click cell A4 (first data row) → View tab → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

When you scroll down, the header row stays visible at the top.

Task 5 3 points

Set document properties: Title and Author

WHY THIS MATTERS

Metadata makes the workbook identifiable and searchable.

WHERE TO FIND IT

File tab → Info → Properties panel. Set Title to “Desert Sun Consulting — Client Billing” and Author to your name.

HOW YOU'LL KNOW IT WORKED

File → Info shows Title and Author populated.

Task 6 3 points

Define a Print Area covering the billing data

WHY THIS MATTERS

A Print Area ensures only the relevant data prints, not blank areas of the worksheet.

WHERE TO FIND IT

Select the billing data range (A1 through the last data row) → Page Layout tab → Print Area → Set Print Area.

HOW YOU'LL KNOW IT WORKED

Print Preview shows only the selected range, not the entire sheet.

Task 7 4 points

Insert a hyperlink to the company website

WHY THIS MATTERS

Hyperlinks let users click directly to an external resource.

WHERE TO FIND IT

Click in an empty cell → Insert tab → Link → type a URL (like https://www.desertsunconsulting.com) → OK.

HOW YOU'LL KNOW IT WORKED

The cell shows blue underlined text; Ctrl+click opens the URL.

D2 — Cell Formatting 19 points · 6 tasks

Task 8 3 points

Merge and center the title row (A1:F1)

WHY THIS MATTERS

Merging creates a wide banner for the workbook title across the top of the data.

WHERE TO FIND IT

Select cells A1 through F1 → Home tab → Alignment group → Merge & Center.

HOW YOU'LL KNOW IT WORKED

The title spans all six columns in one centered cell.

Task 9 3 points

Center-align the column headers

WHY THIS MATTERS

Centered headers look professional and align with the data below.

WHERE TO FIND IT

Select the header row cells → Home tab → Alignment group → Center.

HOW YOU'LL KNOW IT WORKED

All column headers are centered in their cells.

Task 10 3 points

Apply Currency format to the Amount Billed column

WHY THIS MATTERS

Currency formatting adds dollar signs and two decimal places to financial data.

WHERE TO FIND IT

Select the Amount Billed column data → Home tab → Number group → Currency (or Ctrl+Shift+4).

HOW YOU'LL KNOW IT WORKED

Values show as $12,500.00 format.

Task 11 3 points

Apply bold and a fill color to the header row

WHY THIS MATTERS

Bold text and a colored background make the header row stand out from the data rows.

WHERE TO FIND IT

Select the header row → Home tab → Font group → Bold (Ctrl+B) → then click the Fill Color button and choose a color.

HOW YOU'LL KNOW IT WORKED

The header row has bold text on a colored background.

Task 12 3 points

Enable Wrap Text on the header row

WHY THIS MATTERS

Wrap Text prevents long column headers from being cut off or overlapping into adjacent cells.

WHERE TO FIND IT

Select the header row → Home tab → Alignment group → Wrap Text.

HOW YOU'LL KNOW IT WORKED

Long headers wrap to multiple lines within their cells instead of being cut off.

Task 13 4 points

Create a named range called “Billing”

WHY THIS MATTERS

Named ranges make formulas easier to read and data easier to reference.

WHERE TO FIND IT

Select the billing data range → click in the Name Box (left of the formula bar) → type Billing → press Enter.

HOW YOU'LL KNOW IT WORKED

The Name Box shows “Billing” when the range is selected; Formulas → Name Manager lists it.

D3 — Conditional Formatting & Sparklines 10 points · 2 tasks

Task 14 5 points

Apply conditional formatting to highlight values

WHY THIS MATTERS

Conditional formatting visually flags important data — like high or overdue amounts.

WHERE TO FIND IT

Select the Amount Billed column → Home tab → Conditional Formatting → Highlight Cell Rules → Greater Than → enter a threshold value → choose a format → OK.

HOW YOU'LL KNOW IT WORKED

Cells meeting your condition are highlighted with a color or format.

Task 15 5 points

Insert sparklines to show billing trends

WHY THIS MATTERS

Sparklines are tiny in-cell charts that show trends at a glance.

WHERE TO FIND IT

Click in an empty cell next to the data → Insert tab → Sparklines group → Line → select the data range → OK.

HOW YOU'LL KNOW IT WORKED

A small line chart appears inside the cell showing the data trend.

D4 — Tables & Data Tools 21 points · 5 tasks

Task 16 5 points

Convert the billing data to an Excel Table named “BillingTable”

WHY THIS MATTERS

Excel Tables add automatic filtering, structured references, and easy formatting. The table must be named exactly “BillingTable” for the grader to detect it.

WHERE TO FIND IT

Click anywhere in the billing data (A3:H12) → press Ctrl+T (or Insert tab → Table) → verify the range is correct → OK. Then click the Table Design tab → in the Table Name box (top left), change the name to BillingTable → press Enter.

HOW YOU'LL KNOW IT WORKED

Clicking in the data shows the Table Design tab; the Table Name box reads “BillingTable.”

Task 17 3 points

Apply a Medium table style

WHY THIS MATTERS

Table styles add professional banded rows, header shading, and borders in one click.

WHERE TO FIND IT

Click in the table → Table Design tab → Table Styles gallery → choose any Medium style.

HOW YOU'LL KNOW IT WORKED

The table has alternating row colors and a styled header.

Task 18 5 points

Add a Total Row summing the Amount Billed column

WHY THIS MATTERS

The Total Row uses the built-in table feature — not a manual SUM formula. It must be enabled through the Table Design tab.

WHERE TO FIND IT

Click anywhere in BillingTable → Table Design tab → check the Total Row checkbox. Then click in the Total Row cell under the Amount Billed column → click the dropdown arrow that appears → select Sum.

HOW YOU'LL KNOW IT WORKED

A Total Row appears at the bottom of the table with the sum of Amount Billed. The Table Design tab shows Total Row checked.

Task 19 5 points

Filter the table to show only Active clients

WHY THIS MATTERS

Filtering hides rows that don’t match your criteria, showing only the data you need.

WHERE TO FIND IT

Click the dropdown arrow on the Status column header → uncheck “Select All” → check only “Active” → OK.

HOW YOU'LL KNOW IT WORKED

Only rows with Status = Active are visible; the Status dropdown arrow shows a filter icon.

Task 20 3 points

Sort by Client name then by Invoice Date

WHY THIS MATTERS

Multi-level sorting organizes data by a primary field and breaks ties with a secondary field.

WHERE TO FIND IT

Data tab → Sort → Sort by: Client → Add Level → Then by: Invoice Date → OK.

HOW YOU'LL KNOW IT WORKED

Data is sorted alphabetically by Client; within each client, rows are sorted by date.

D5 — Charts 21 points · 4 tasks

Task 21 5 points

Create a Column chart comparing revenue by client

WHY THIS MATTERS

Charts visualize billing data and make comparisons easy to see at a glance.

WHERE TO FIND IT

Select the Client names and Amount Billed columns (including headers) → Insert tab → Charts group → Insert Column Chart → Clustered Column.

HOW YOU'LL KNOW IT WORKED

A column chart appears showing billing amounts for each client.

Task 22 6 points

Add a second data series for Prior Year comparison

WHY THIS MATTERS

A two-series chart lets you compare current billing to last year. The Prior Year data is already in column G of the Billing sheet — you need to add it to the chart.

WHERE TO FIND IT

Right-click the chart → Select Data → Add → Series name: click the “Prior Year” header (G3) → Series values: select G4:G11 → OK.

HOW YOU'LL KNOW IT WORKED

The chart shows two sets of columns side by side — current year and prior year — for each client.

Task 23 5 points

Add data labels and an axis title to the chart

WHY THIS MATTERS

Data labels show the exact value on each column so readers don’t have to estimate from the axis, and an axis title explains what the numbers measure.

WHERE TO FIND IT

Click the chart → click the Chart Elements button (the + beside the top-right corner of the chart) → check Data Labels → check Axis Titles → click the vertical axis title placeholder and type “Amount Billed ($)”. (You can also use Chart Design tab → Add Chart Element → Data Labels, then Add Chart Element → Axis Titles.)

HOW YOU'LL KNOW IT WORKED

A value appears on each column, and the vertical axis shows your title. Both boxes are checked in the Chart Elements list.

Task 24 5 points

Apply a chart style

WHY THIS MATTERS

Chart styles apply a coordinated set of colors, fonts, and effects in one click, giving the chart a polished, professional look.

WHERE TO FIND IT

Click the chart → Chart Design tab → Chart Styles gallery → click any style other than the one already applied (for example, Style 6). You can also click the Chart Styles button (the paintbrush beside the chart).

HOW YOU'LL KNOW IT WORKED

The chart's look changes, and the style you picked is highlighted in the Chart Styles gallery.

D6 — Accessibility 4 points · 1 task

Task 25 4 points

Add alt text to the chart

WHY THIS MATTERS

Alt text describes visuals for screen readers and is important for accessibility compliance.

WHERE TO FIND IT

Right-click the chart → Edit Alt Text → type a clear description of what the chart shows (e.g., "Column chart comparing current and prior year billing by client").

HOW YOU'LL KNOW IT WORKED

Right-clicking the chart and selecting Edit Alt Text shows your description.

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